

Early *Candida* Isolation and Long-Term Survival After Lung Transplantation: Species-Specific Risks and Predictors

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INTRODUCTION

BACKGROUND

- Candida* species are frequent nosocomial and donor-derived fungal pathogens among lung transplant recipients
- Despite their prevalence, the impact of early *Candida* isolation on post-transplant outcomes remains poorly defined.
- Species-specific variation in pathogenicity may influence clinical course and outcomes, yet comparative data are limited.
- With evolving fungal epidemiology in transplant populations, clarifying the prognostic significance and predictors of *Candida* infection is essential to optimize management and improve survival.

OBJECTIVE:

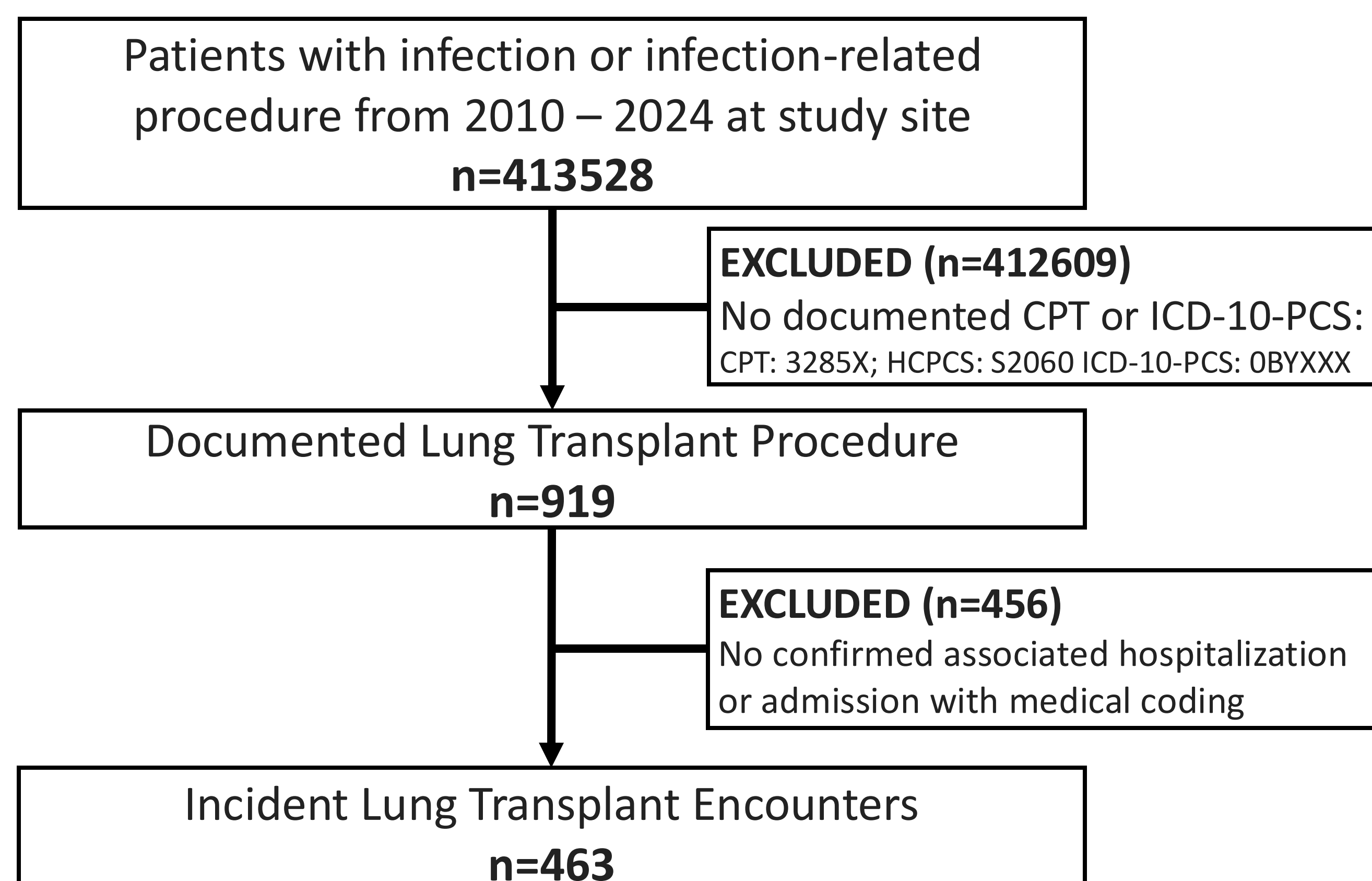
- To evaluate the association between early *Candida* isolation and post-transplant survival, and to identify clinical predictors of infection in lung transplant recipients.

METHODS

STUDY DESIGN

- Retrospective cohort of incident lung transplant cases using electronic health records from 2010 to 2024 at a tertiary-care hospital
- Eligibility:** Lung transplant procedural billing coding (ICD, CPT) with a confirmed associated hospitalization (**Figure 1**)
- Exposures:** *Candida* isolated within 30 days after transplant
- Outcomes:** Overall survival at 1, 5, and 10 years.
- Analysis:** Cohort was summarized descriptively (**Table 1, Figure 2**). Adjusted Cox models assessed associations between *Candida* isolation and survival (**Figure 3**). Multivariable logistic regression identified pre-transplant predictors of *Candida* isolation (**Figure 4**).

STUDY SCHEMA



RESULTS OVERVIEW: TABLES & FIGURES

Table 1: Descriptive Summary of Study Cohort

Characteristic	Overall (n = 463)
Follow-Up (Years) - Median (Min,Max)	5.3 (0, 14)
Age at Transplant - Years (SD)	56.0 (14.3)
Sex (Female) - n (%)	176 (38%)
Race (White) - n (%)	376 (81.2%)
Ethnicity (Hispanic) - n (%)	71 (15.3%)
Transplant Type - n (%)	
Double Lung	395 (85.3%)
Single Lung	68 (14.7%)
Year of Transplant - n (%)	
2010-2014	65 (14%)
2015-2019	184 (39.7%)
2020+	214 (46.2%)
Prior Comorbidities – n (%)	
Chronic Pulmonary Disease	416 (89.8%)
Peripheral Vascular Disease	287 (62%)
Congestive Heart Failure	190 (41%)
Cerebrovascular Disease	121 (26.1%)
Diabetes w/o complications	109 (23.5%)
Renal disease	103 (22.2%)
Mild Liver Disease	93 (20.1%)
Overall Survival (Adjusted) – %	
1-Year Survival	92.5%
5-Year Survival	66.8%
10-Year Survival	42.0%

Figure 2: Distribution of *Candida* species Isolated Within 30 Days Post-Lung transplant.

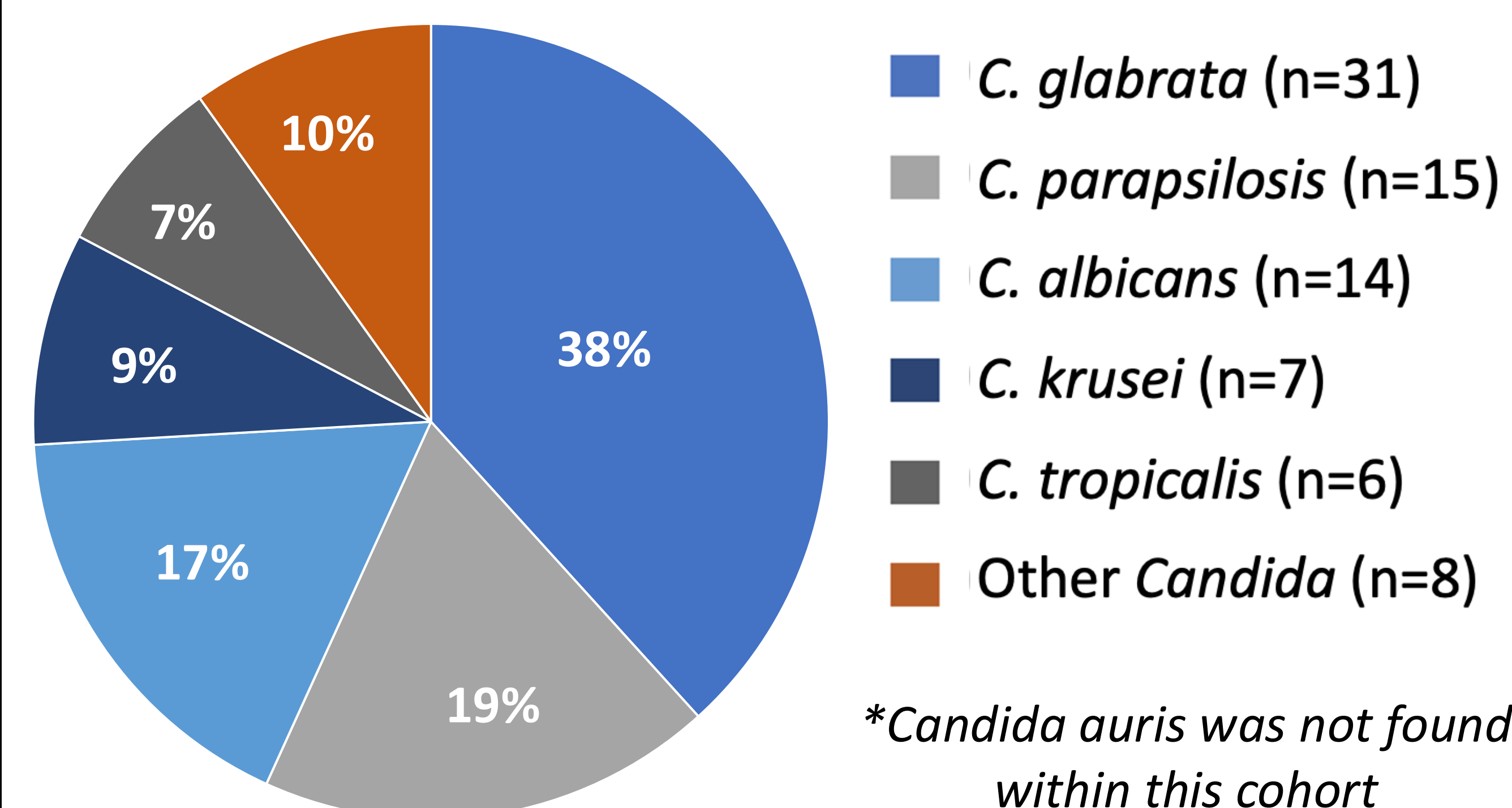


Figure 3: Adjusted Survival Curves by Early Post-Transplant *Candida* isolation

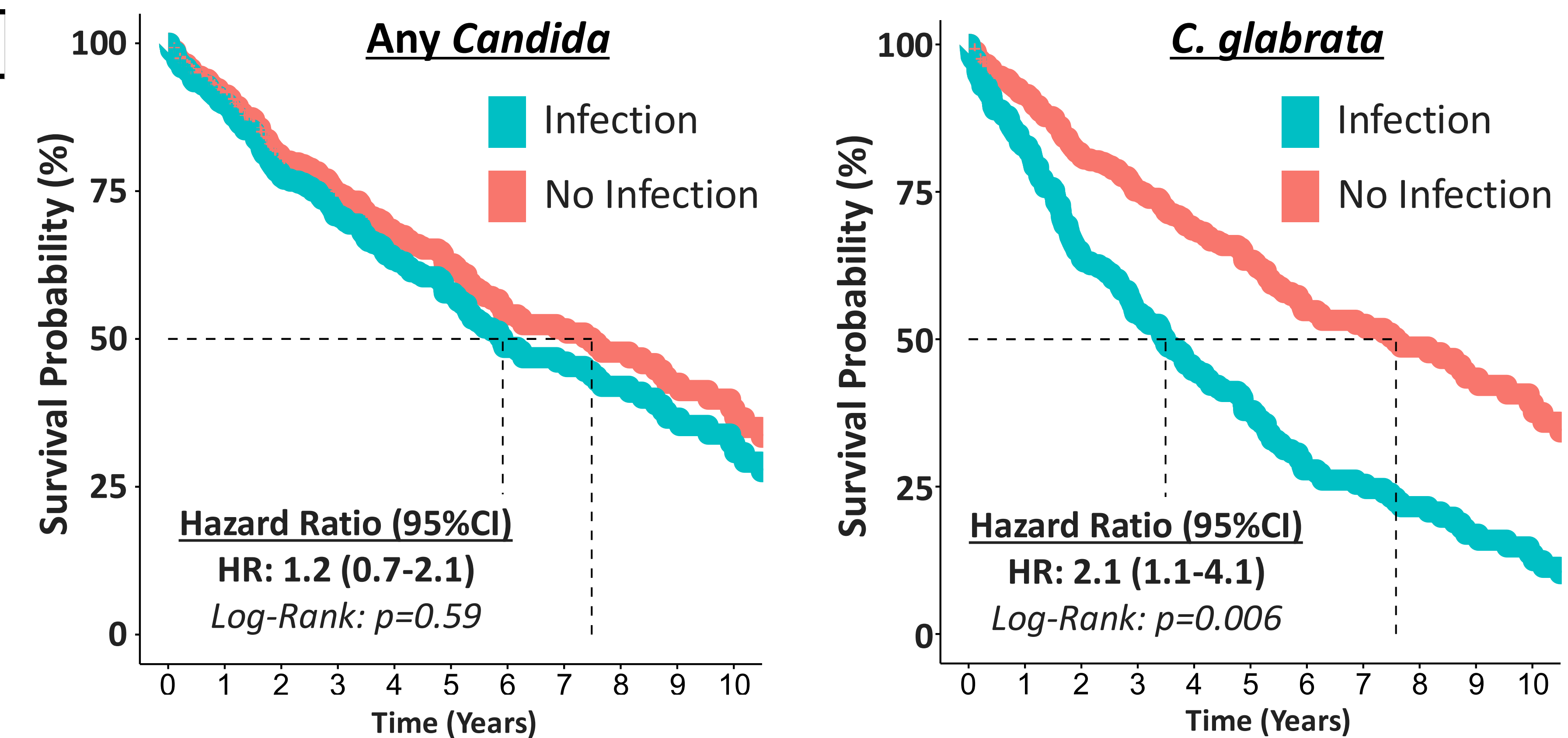
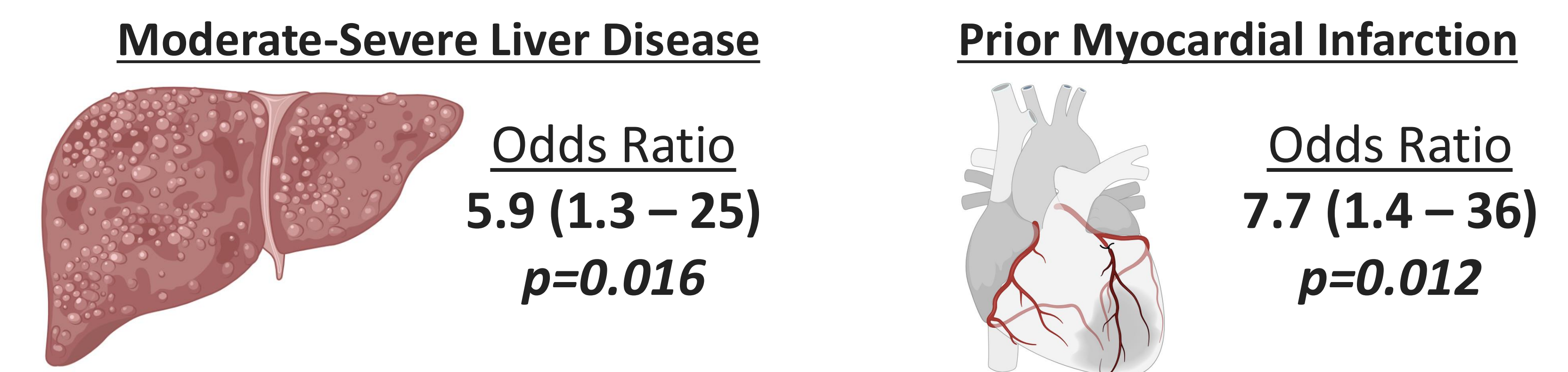


Figure 4: Pre-Transplant Predictors of Early *C. glabrata* Isolation



*Adjusted for age, sex, race, ethnicity, comorbidities, other transplant-specific factors

CONCLUSION

- In this exploratory analysis, early *Candida* isolation, particularly *C. glabrata*, was associated with reduced post-transplant survival among lung transplant recipients.
- Prior liver disease and myocardial infarction, were significant predictors of *C. glabrata* risk, highlighting the potential for targeted risk stratification.
- Future studies will expand the study cohort and assess the effectiveness of antifungal prophylaxis strategies in improving clinical outcomes and survival.

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